First part of our project:

1. Data preparation and Train-Test split

We merged the publicly available World Happiness Report datasets from **2015 to 2023**. Before fitting the regression model, several data cleaning steps were performed to ensure consistency across years and to remove missing or unusable observations. This resulted in **1367 complete country-year observations**. We selected 2015-2022 data as the training set and the 2023 data as the testing set

2. Regression model

We first fit a standard multiple linear regression model on the training data:

$$\text{Happiness}_i = \beta_0 + \beta_1 \text{GDP}_i + \beta_2 \text{SocialSupport}_i + \beta_3 \text{LifeExpectancy}_i$$
$$+ \beta_4 \text{Freedom}_i + \beta_5 \text{Generosity}_i + \beta_6 \text{Corruption}_i + \varepsilon_i.$$

Result:
GDP β_1=0.904: Higher income levels are associated with higher happiness, reflecting the role of economic development in subjective well-being.

Social support β_2=0.738: Stronger social networks significantly improve national happiness levels.

Healthy life expectancy β_3=1.151: Health is one of the strongest predictors; countries with longer healthy lifespans tend to be notably happier.

Freedom β_4=1.567: This is the largest coefficient in the model, showing that perceived freedom dramatically enhances subjective well-being.

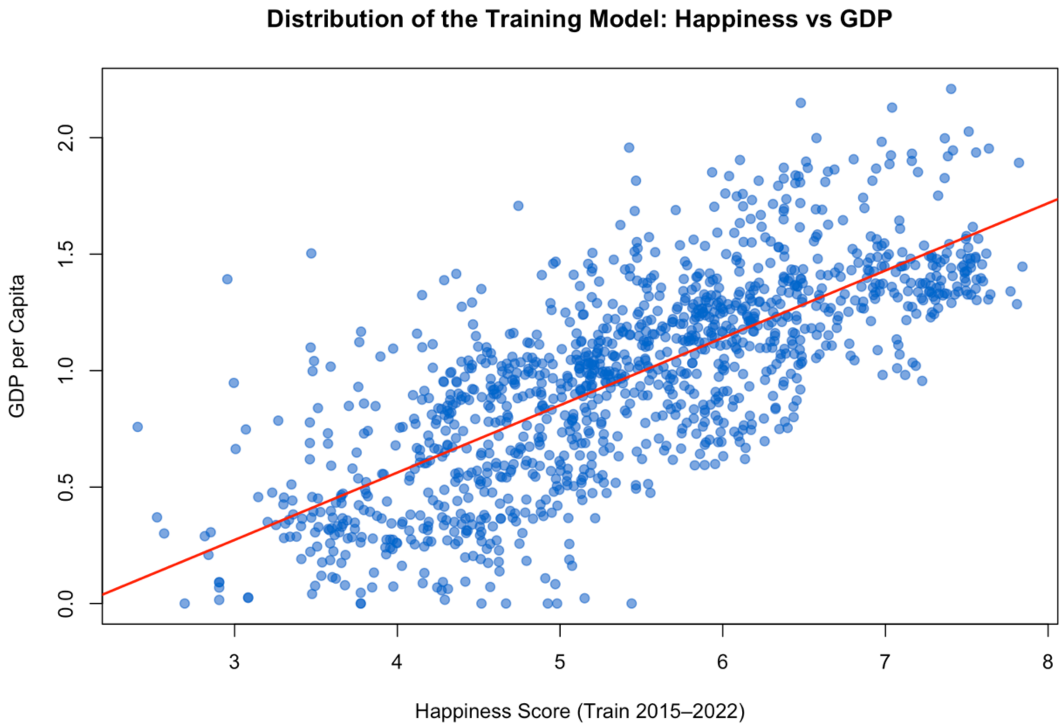
Generosity β_5=0.699: More generous societies tend to report higher happiness.

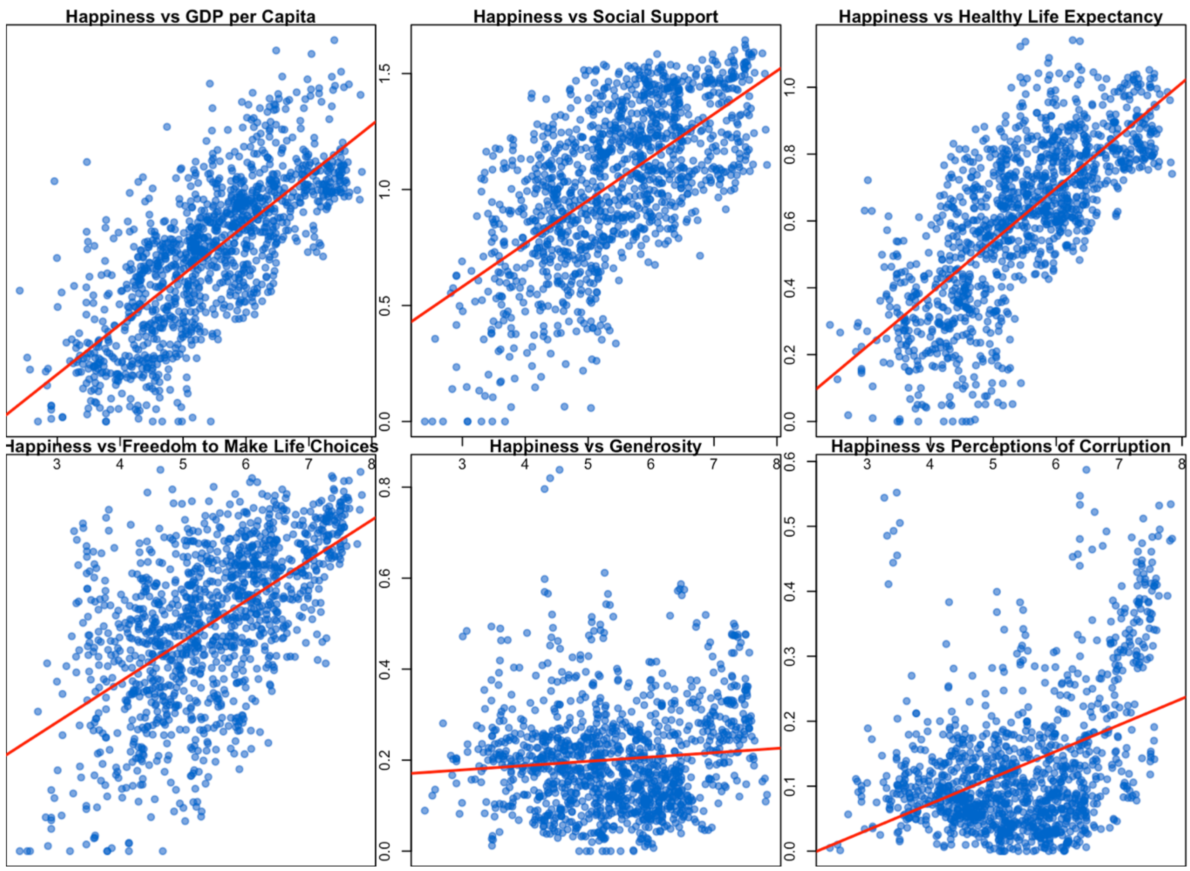
Corruption β_6=0.821: Lower corruption is strongly associated with higher happiness.

The model achieves an adjusted R^2 of 0.7415. This level of fit is considered strong for cross-country social science data, where human well-being is affected by numerous unobserved cultural, historical, and institutional factors.

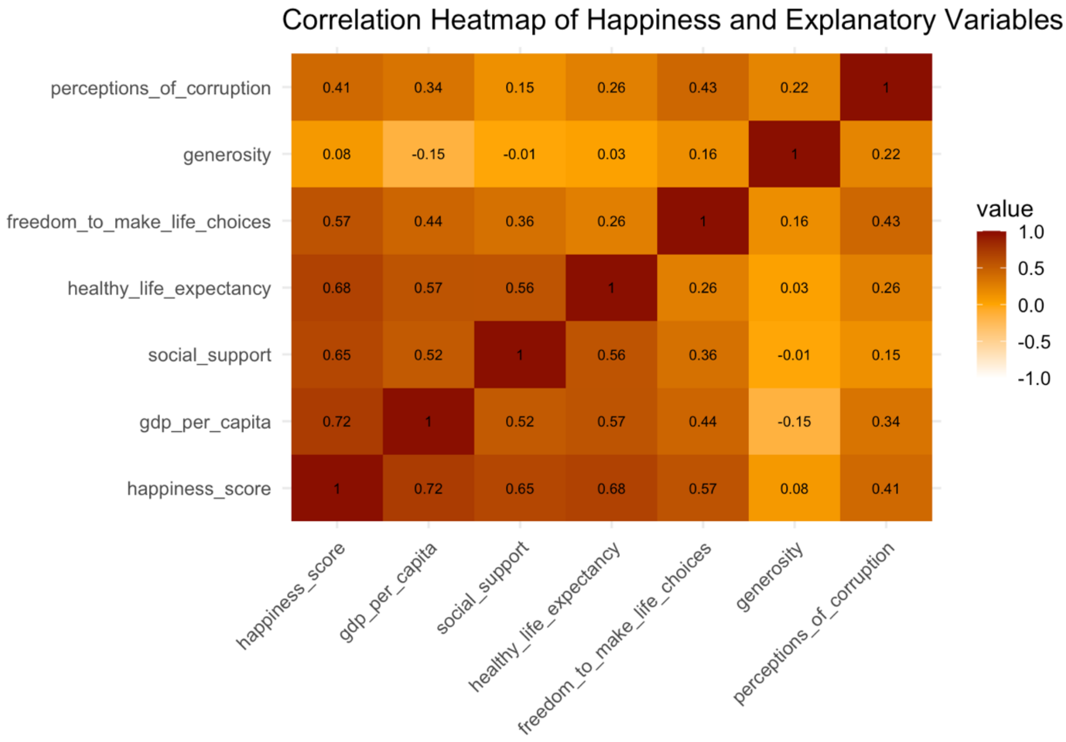
All six explanatory variables are statistically significant at the $p < 0.001$ level, suggesting that each factor contributes meaningfully to the prediction of the happiness score.

The resulting model is shown in the figures below:


Distribution of the Training Model: Happiness vs GDP
GDP per Capita
Happiness Score (Train 2015–2022)


Happiness vs GDP per Capita
Happiness vs Social Support
Happiness vs Healthy Life Expectancy
Happiness vs Freedom to Make Life Choices
Happiness vs Generosity
Happiness vs Perceptions of Corruption

The heat map of each parameter:



3. Prediction performance

RMSE=0.551

MSE=0.304

MAE=0.394

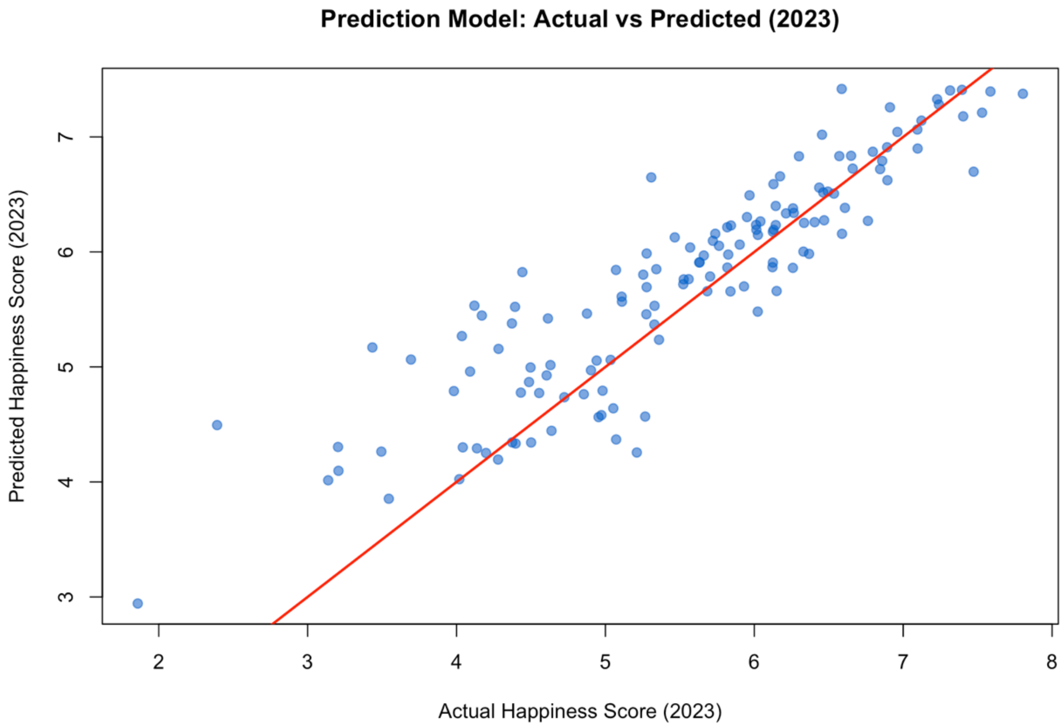
Given that the happiness score typically ranges from 3 to 8 for most countries, these results indicate that the model's predictions deviate from the true values by approximately 0.4–0.6 points on average.

$$MAE = \frac{1}{m}\sum_{i=1}^{m} | y_i - y_i^{'} |^2$$

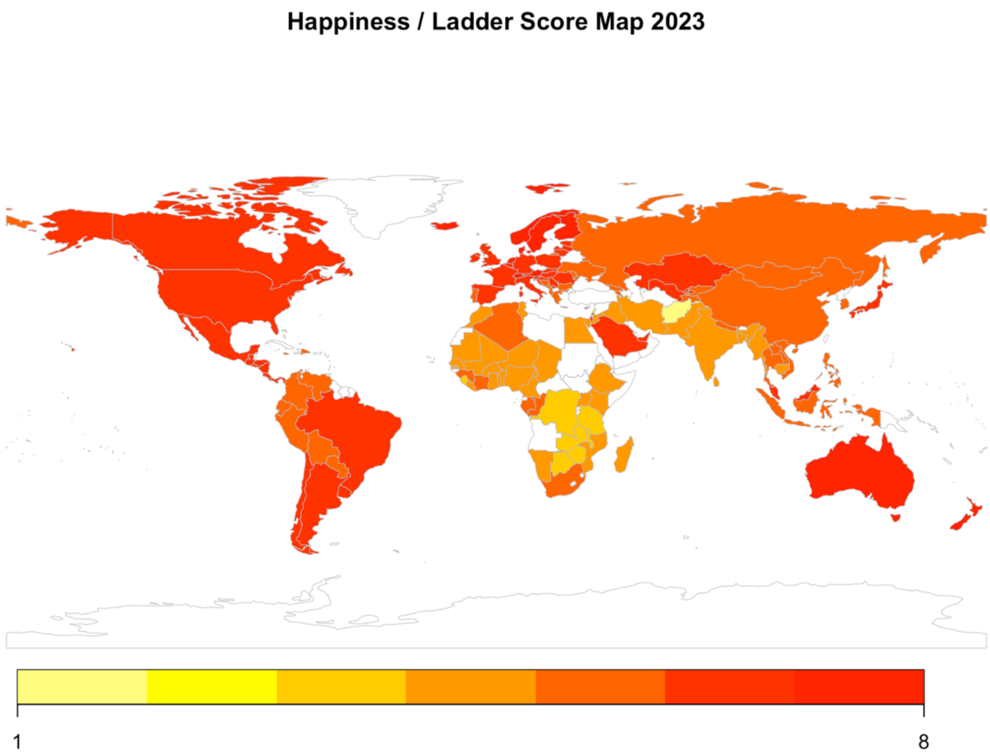
$$MSE = \frac{1}{m}\sum_{i=1}^{m} \left(y_i - y_i^{'} \right)^2$$

$$RMSE = \sqrt{MSE}$$

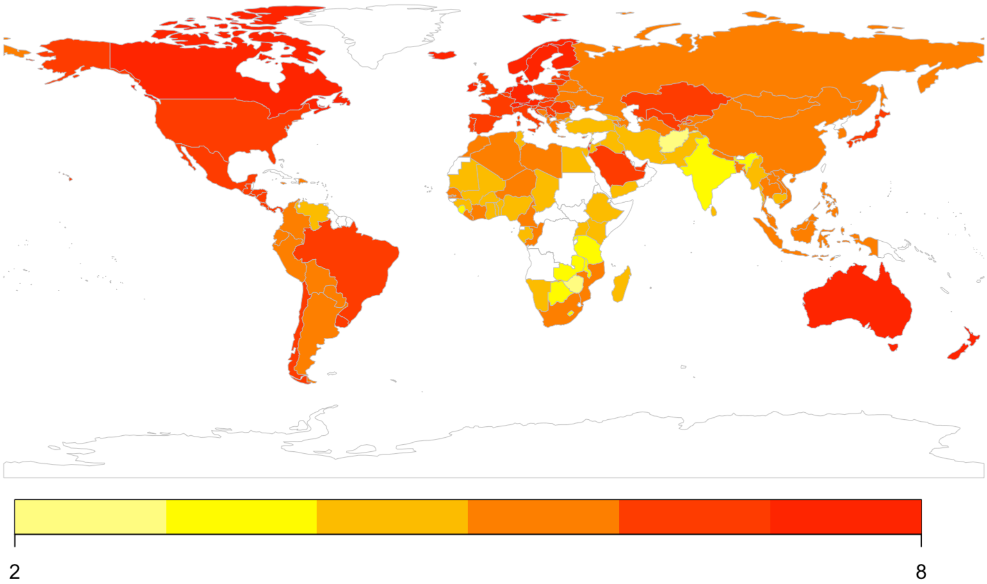
Prediction model figure:



2022 and 2023 Happiness score distribution map:



Happiness / Ladder Score Map, 2022



Predicted average happiness scores across major world regions in 2023 (including error)